Must Read Before Use

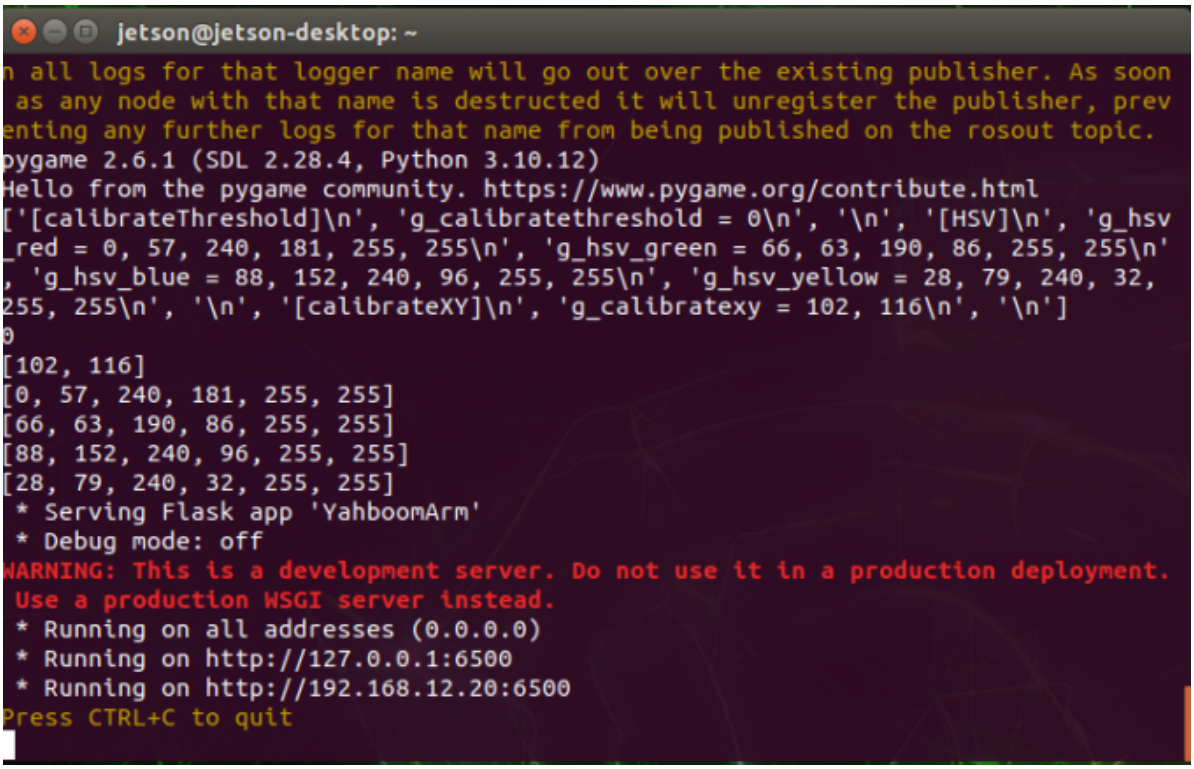
1. Entering Docker Container

1.1. First Entry

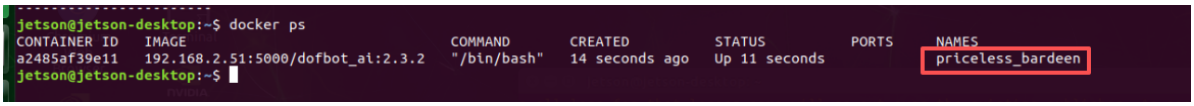
After the robotic arm boots up and enters the desktop normally, log in to the mainboard via SSH or VNC, open a terminal, and enter the following command to start a new Docker container:

```
sh Docker_ROS.sh
```

This will enter the container. On first entry, the APP main program control will start by default. You don't need to close this for now.



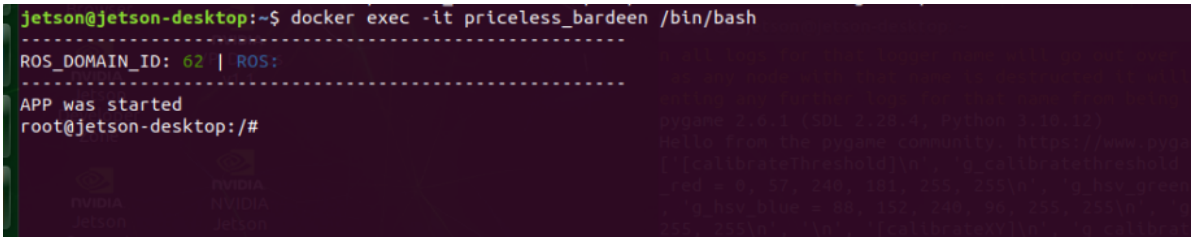
Next, open another terminal on the mainboard and enter `docker ps` to list the currently running containers, as shown below:



Remember the content in the NAMES column - this is the container name. Use this container name to enter the specified container later. For example, if I want to enter this container, enter the following command in the mainboard terminal:

```
docker exec -it priceless_bardeen /bin/bash
```

Press Enter and you'll be inside:



1.1. Re-entry After Shutdown

If you're restarting, you need to restart the container first before entering. Using priceless_bardeen as an example, enter the following command in the mainboard terminal to restart Docker:

```
docker restart priceless_bardeen
```

Then you can use the exec command to enter Docker. Enter the following command in the mainboard terminal:

```
docker exec -it priceless_bardeen /bin/bash
```

Press Enter and you'll be inside.

2. Calibrating Robotic Arm Offset

Follow these steps strictly.

2.1. Using APP to Calibrate Robotic Arm Center Position

When in the container, if the APP is running, open the mobile APP-YahboomRobot, enter the IP address, connect to the robotic arm, click Robotic Arm Settings - Servo Calibration, and calibrate the robotic arm's center position according to the prompts.

If the APP is not running, enter the following command in the container to start the APP, then perform calibration:

```
sh /root/APP_DOFBOT/start_app.sh
```

2.2. Using Calibration Program to Calibrate Offset

Enter the following command in the container to start the calibration program (first container terminal):

```
ros2 launch dofbot_info camera_arm_kin.launch.py
```

Enter the same container terminal again and input:

```
ros2 run dofbot_sorting get_offset
```

Then fix the robotic arm on the map as shown below:

The program interface will recognize the machine code blocks, then press the spacebar to complete calibration.

Disclaimer: This offset only applies to the current recognition angle. If future program development involves angle modifications, recalibration will be required.